

Cardiometabolic Multiple Long-Term Conditions 2



Biological and mechanistic pathways of cardiometabolic multiple long-term conditions

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Cardiometabolic multiple long-term conditions (MLTC) arise from the complex interplay of biological, sociodemographic, environmental, and behavioural factors across the life course. Shared risk factors and mechanisms, including insulin resistance, adiposity, and chronic inflammation, underpin its development. Growing evidence also implicates that even low-level, long-term exposure to fine particulate matter, nitrogen dioxide, and related pollutants can accelerate the trajectory of cardiometabolic MLTC. Early-life exposures, including undernutrition and overnutrition, altered gut microbiome, and endocrine-disrupting chemicals, interact with social determinants of health to aggravate inflammatory and metabolic dysregulation. These mechanisms, together with genetic susceptibility, epigenetic modifications, and multiomics perturbations, shape disease progression, heterogeneity, and the clustering of cardiometabolic MLTC. Yet, fundamental gaps persist, whereby most mechanistic insights are derived from single-disease studies, leaving the temporal hierarchy, causal pathways, and population-level heterogeneity largely unresolved. Addressing these challenges will require life-course research, integrative systems approaches, and translational studies that link mechanistic insights to precision prevention and therapeutic strategies. By bridging discovery with actions, such efforts can enhance care for cardiometabolic MLTC and promote equitable health outcomes globally.

Introduction

Cardiometabolic multiple long-term conditions (MLTC), most commonly including type 2 diabetes, myocardial infarction, and stroke, have long been recognised as interconnected conditions.^{1,2} Recently, chronic kidney disease (CKD), heart failure, metabolic dysfunction-associated steatotic liver disease (MASLD), hypertension, and atrial fibrillation have been included in the cardiometabolic MLTC spectrum, making it a growing global health challenge with major clinical and policy implications.^{2,3} The first paper in this three-part Series provides a more detailed discussion on the evolving definitions of cardiometabolic MLTC and reviews their epidemiology.² For the purposes of this Series paper, we excluded obesity as a chronic disease in the definition as it is a pathophysiological driver underpinning the development and progression of cardiometabolic MLTC.²

Irrespective of national income levels, the increasing prevalence of cardiometabolic MLTC is mainly driven by an ageing population and shared risk factors, including rapid urbanisation, nutritional transition, smoking, and metabolic risk factors.^{1,3,4} Clustering of these conditions is disproportionately associated with polypharmacy, high rates of health-care use, morbidity, and premature mortality.⁵ Despite its increasing burden, clinical care and research remain siloed within single-disease frameworks, leading to systemic vulnerability, care fragmentation, and adverse health and economic outcomes.

In the second paper in this Series, we examine the biological and mechanistic foundations that link and amplify cardiometabolic MLTC. We synthesise evidence to describe how common perturbations, such as insulin resistance, chronic inflammation, and ectopic fat

deposition, influence disease trajectories. We also highlight emerging insights from genetics, multiomics, and systems biology that reveal early-life determinants, organ crosstalk, and molecular networks as key components of cardiometabolic MLTC. Understanding the mechanistic pathways is essential for reshaping models of prevention, diagnosis, and treatment. Biological mechanisms can characterise the distinct pathways underpinning certain clustering patterns. This enables risk stratification and identifies opportunities for intervention that extend beyond conventional single-disease silos and respond to the dynamic, multisystem

Search strategy and selection criteria

We searched three databases (PubMed, Embase, and Web of Science) from inception to Nov 30, 2025 for articles written in English. We used the search terms “diabetes mellitus”, “cardiovascular disease”, “kidney disease”, “type 2 diabetes”, or “cardiometabolic” alone and in combination with the terms “multimorbidity” or “multiple long-term conditions”. We used the specific MeSH term “cardiometabolic” as an umbrella descriptor for a comprehensive capture of risk factors rather than searching for individual conditions. We selected randomised controlled trials, observational studies (ie, cross-sectional, retrospective, and prospective cohort), and systematic reviews and meta-analyses from the past 10 years, but did not exclude commonly referenced and highly regarded older publications. We also searched the reference lists of the articles identified and selected those we judged to be relevant. Review articles are cited to provide readers with more details and references beyond the scope of this Series paper.

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This is the second in a [Series](#) of three papers about cardiometabolic multiple long-term conditions. All papers in the Series are available at www.thelancet.com/series-do/cardiometabolic-mltc

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Key messages

- Cardiometabolic multiple long-term conditions (MLTC) are underpinned by shared, interconnected biological mechanisms. The central pathways driving organ dysfunctions include insulin resistance, ectopic fat deposition, chronic inflammation, and increased oxidative stress.
- Social determinants of health modulate systemic vulnerability and substantially influence the trajectory of cardiometabolic MLTC. Environmental factors, including pollution, accelerate disease progression.
- Nutritional transitions (undernutrition and overnutrition), gut microbiome dysbiosis, and endocrine-disrupting chemicals interact to worsen metabolic dysregulation. These early-life exposures imprint durable genetic and epigenetic effects, establishing a lifelong susceptibility to cardiometabolic MLTC.
- Leveraging genetic data through polygenic risk scores can improve population risk prediction, enable genomic risk stratification, and validate mechanistic targets for therapeutic development of cardiometabolic MLTC.
- High-throughput multiomics profiling serves as an observation window into shared biological mechanisms. Integrating these emerging biomarkers improves predictive performance for incident cardiometabolic MLTC beyond established clinical models.
- Addressing the growing burden of MLTC requires moving away from siloed, single-disease clinical frameworks. Future research and management strategies must embrace life-course cohort studies, diverse cross-ancestral data, and systems biology to drive precision prevention and treatment.

nature of cardiometabolic MLTC. For clarity throughout this Series paper, MLTC refers to multiple long-term conditions broadly, which is distinct from our specific focus on cardiometabolic MLTC.

By integrating foundational biology with emerging mechanistic evidence, this review provides the scientific basis needed to inform the evolving management and therapeutic strategies (discussed in the third paper in the Series)² and the epidemiological landscape and unmet needs of cardiometabolic MLTC (discussed in the first paper in the Series).⁶

Shared risk factors and mechanisms

There are common risk factors for cardiometabolic conditions, including dysglycaemia, hypertension, adiposity (including visceral fat), and smoking. Attention to these and other modulators, such as social determinants of health (SDOH) and environmental factors, can help mitigate adverse outcomes. A greater understanding of and, where possible, treatment of pathogenic factors is desirable. Basic science is helping to inform the underlying pathophysiology, which will guide the

development and use of new or repurposed treatments and health policy. Core mechanisms underlying organ dysfunction and damage include insulin resistance, ectopic fat deposition (including visceral fat), chronic inflammation, and increased oxidative stress.⁷

Oxidative stress involves mitochondrial dysfunction and drives the formation of advanced glycation end products produced through the interactions of glucose, proteins, and lipids.^{8,9} There is multiorgan (eg, adipose tissue, blood vessels, myocardium, kidneys, liver, pancreas, and skeletal muscle) crosstalk and many self-perpetuating loops or vicious cycles. Figure 1 shows the commonality of environmental, social, and genetic factors and related epigenetic changes underlying cardiometabolic MLTC, and the key mechanisms are outlined later in this paper. Discerning causation from association can be difficult. Challenges relate to the common coexistence of multiple risk factors and comorbidities, epigenetic and metabolic memory, bidirectional feedback loops, and a lack of comprehensive life-course studies with detailed genotyping and phenotyping.

Insulin resistance

Insulin resistance arises when normally insulin-sensitive tissues, such as adipose tissue, liver, and skeletal muscle, become less responsive to insulin produced by pancreatic β cells. This core metabolic defect drives hyperglycaemia and promotes atherogenic dyslipidaemia, hypertension, and kidney injury, linking type 2 diabetes with cardiovascular disease and CKD. Diabetes is defined and diagnosed based on hyperglycaemia, but insulin also has major roles in protein and lipid metabolism. Insulin modulates cell signalling, energy homeostasis, and appetite, is vasoactive, and is a growth factor.¹⁰ Key sites of insulin action include white adipose tissue, liver, skeletal muscle, and the brain.¹⁰

The insulin signalling pathway comprises proximal canonical elements of the insulin receptor, insulin receptor substrate proteins, PI3K, and AKT, with many bidirectional regulatory feedback loops. Distal elements are substrates of AKT with multiple activating phosphorylation sites.¹¹ Notably, insulin signalling pathways are linked with AMPK—a cellular energy sensor. AMPK activation promotes glucose uptake, increases insulin sensitivity, inhibits β -cell stress and death, and thus, reduces the risk of diabetes-related end-organ damage. AMPK signalling interacts with other key molecular pathways, including PI3K–Akt, NOX4, NF- κ B, and PGC-1 α , which are involved in inflammation, oxidative stress, cellular metabolism, proliferation, and survival.^{11,12} Disruption of insulin and AMPK signalling further worsens glucose uptake and increases hepatic glucose production, perpetuating hyperglycaemia, insulin resistance, and lipotoxicity, leading to β -cell failure in insulin production and secretion.^{11,12} These changes drive hepatic production of triglyceride-rich

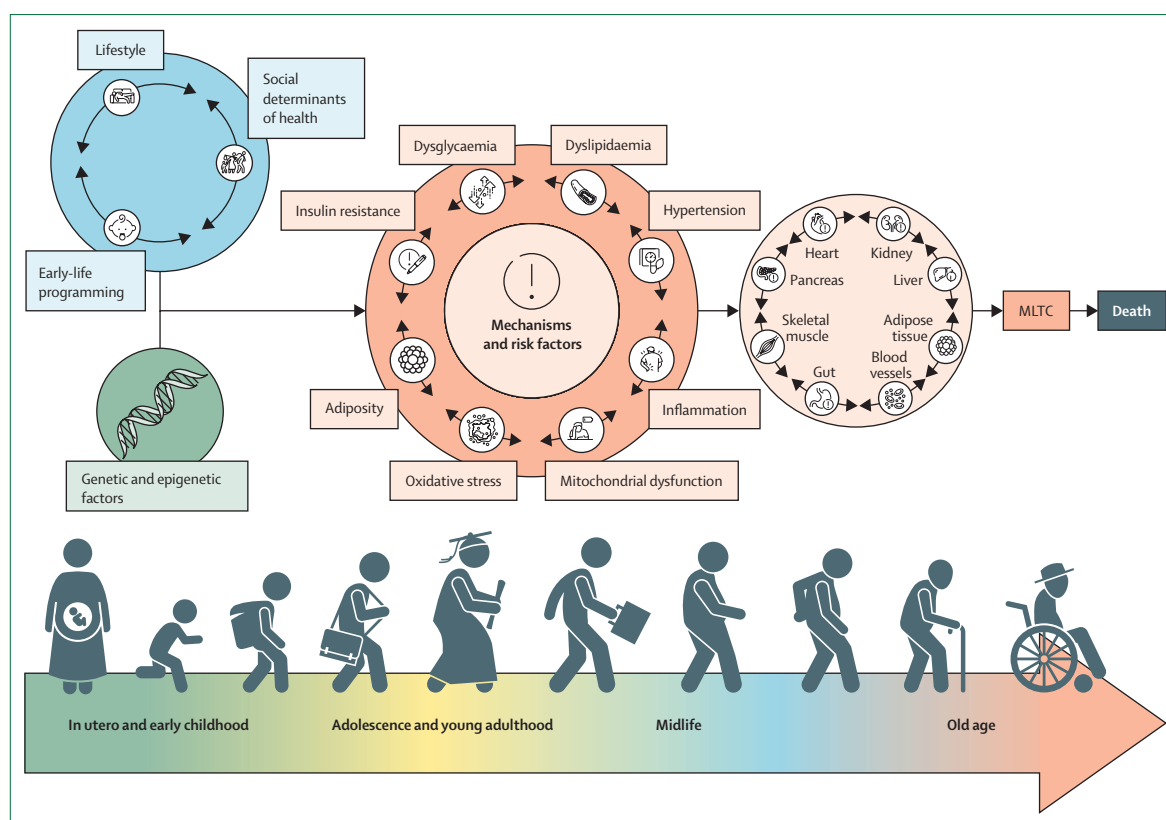


Figure 1: An integrated model illustrating the common mechanisms of how social, behavioural, environmental, genetic, and epigenetic factors interact across the life course in the development and progression of cardiometabolic MLTC
MLTC=multiple long-term conditions.

VLDLs, the breakdown products of which (ie, VLDL remnants and LDLs) are proatherogenic.

Notably, insulin resistance and low insulin levels affect the activity of key enzymes in lipoprotein metabolism, namely lipoprotein lipase on the vascular endothelium and hepatic lipase in the liver, thereby reducing VLDL catabolism and HDL synthesis.^{13,14} These perturbations lead to high triglyceride and low HDL cholesterol levels, which are common in people with cardiovascular–kidney–metabolic conditions.^{13,14} Low HDL cholesterol is both a cause and a consequence of impaired insulin secretion. HDL cholesterol protects against β -cell apoptosis, promotes insulin secretion, and has anti-inflammatory, antioxidant, vasodilatory, and growth-promoting actions.¹³

Hypertriglyceridaemia and ectopic fat deposition

Hypertriglyceridaemia is similarly pathogenic, being associated with increased skeletal muscle triglyceride and fat accumulation, which further worsens insulin resistance.¹⁵ Increased free fatty acid levels and hyperlipidaemia drive ectopic lipid deposition in the liver (contributing to MASLD), as well as in the myocardium, blood vessels, and kidneys, thereby promoting organ dysfunction.⁷ Alterations in adipose

tissue and hepatic metabolism further lead to dysregulated secretion of adipokines and cytokines, including adiponectin and resistin (which modulate insulin sensitivity) and proinflammatory markers, including tumour necrosis factor (TNF) and interleukin (IL)-6, IL-1 β , and IL-18.^{16,17} TNF also stimulates hepatic production of C-reactive protein—an inflammatory marker linked to adiposity, type 2 diabetes, cardiovascular disease, and CKD.¹⁸

Apart from its inflammatory and endocrine functions, visceral fat surrounding organs has local mechanical effects.^{19–21} Epicardial fat can reduce diastolic function, contributing to heart failure with preserved ejection fraction.^{20,21} Excess visceral fat exerts pressure on the abdomen and diaphragm, which can increase breathlessness, risk of sleep apnoea, and reduced blood flow to vital organs.¹⁹

Chronic inflammation

Even low-grade chronic inflammation exacerbates insulin resistance by disrupting intracellular signalling and activating other inflammatory pathways, including profibrotic pathways such as the NLRP3 inflammasome, which contributes to cardiovascular–kidney–liver

injury.^{18,22,23} This inflammatory milieu also drives endothelial dysfunction and vascular inflammation, reflected by elevated vascular cell adhesion molecules, alongside vasoconstriction and increased vascular permeability arising from imbalances between dilatory and constricting factors such as nitric oxide and endothelin-1. Inflammation also promotes a prothrombotic state due to dysregulation of platelet activity, fibrinogen, coagulation factors, and fibrinolytic mediators (including tissue plasminogen activators and plasminogen activator inhibitors), many of which are synthesised by the liver.

Oxidative stress

An imbalance between the production of reactive oxygen species and extracellular and intracellular antioxidant defences leads to oxidative damage of lipids, proteins, DNA, cells, and tissues. SDOH and lifestyle factors can contribute to oxidative stress, such as through pollution, smoking, and a poor-quality diet, as discussed below. Oxidative stress also enhances inflammation and insulin resistance and accelerates the formation of advanced glycation end products, which interact with their receptors to worsen tissue injury. Advanced glycation end products form rapidly within cells and more slowly in blood vessels and are markedly increased in diabetes and CKD. Their accumulation contributes to vascular and myocardial stiffness, kidney fibrosis and dysfunction, and liver disease.^{8,9,24}

Mitochondrial dysfunction

Mitochondria are strongly linked to oxidative stress, but they also have roles in cell signalling, tissue metabolism, tissue development, neurodegeneration, immunity, and preventing infectious diseases. Studies on genetic mitochondrial disorders also support their key roles in skeletal muscle, the heart, the brain, the liver, and the pancreas.^{25,26}

Central to Brownlee's unifying hypothesis of diabetes-related complications is mitochondrial dysfunction,²⁷ which is also implicated across other cardiometabolic and kidney conditions.^{27–29} Hyperglycaemia promotes synthesis of mitochondrial reactive oxygen species, activating four major pathogenic pathways, namely the formation of advanced glycation end products, PKC, and the polyol and hexosamine pathways.²⁷ The mechanistic interconnectivity of these pathways via mitochondrial dysfunction might explain why clinical trials targeting individual pathways, such as PKC inhibition, have not improved primary endpoints, whereas experimental models show that attenuating mitochondrial reactive oxygen species overproduction mitigates cellular and tissue injury.²⁷ Interestingly, a xenomitochondrial mouse model with an altered mitochondrial respiratory chain exhibits an increase in oxidative stress and myocardial dysfunction, and perpetuates damage in the myocardium, kidney, and blood vessels when diabetes is

induced.³⁰ This model offers a valuable platform for testing mitochondrial-targeted interventions.

Other pathways

Apart from the core mechanisms of insulin resistance, ectopic fat deposition, and chronic inflammation, other pathways implicated in the pathogenesis of cardiometabolic MLTC are activation of the renin–angiotensin system, alterations in the gut microbiome, and genetic susceptibility.^{1,31} Activation of the renin–angiotensin system promotes endothelial dysfunction, hypertension, fibrosis, and impairs vascular remodelling, leading to damage of the myocardium, kidney, and liver. Angiotensin II—a key renin–angiotensin system effector—drives oxidative stress and activates hepatic macrophages, thereby promoting the development of MASLD.^{32–34} To date, no single genetic factor has been identified for cardiometabolic MLTC, although ongoing research continues to elucidate loci associated with individual conditions.

Although these shared risk factors and mechanisms offer a coherent framework, they do not fully account for the disease heterogeneity and complexity across populations with cardiometabolic MLTC. Hence, the following sections explore emerging mechanistic insights, spanning from environmental and psychosocial factors (eg, early-life exposures and SDOH) to molecular signatures that collectively advance our understanding of the development of cardiometabolic MLTC.

Role of SDOH

Although shared biological pathways shape the clustering and progression of cardiometabolic MLTC, these pathways operate within broader sociocultural contexts. SDOH influence exposure to these risks, modulate systemic vulnerability, and shape the trajectory of cardiometabolic MLTC. Adverse SDOH also contribute to other long-term conditions in addition to cardiometabolic conditions, including mental health and cognitive conditions.³⁵ This section focuses on how SDOH interact with mechanistic pathways to drive the burden of cardiometabolic MLTC (figure 1).

SDOH domains and disease risk

Six core domains of SDOH have been identified as key drivers of risk and progression of cardiometabolic MLTC. These domains include (1) economic stability, including income, employment, and financial security; (2) education, including educational attainment and literacy; (3) health-care access and quality, including insurance coverage and availability of providers; (4) food security, including consistent access to nutritious food; (5) neighbourhood and built environment, including housing quality, safety, and environmental exposures; and (6) social and community context, including support, discrimination, and social capital.^{36–39}

Growing evidence indicates that SDOH have a large effect on cardiometabolic disease risk. People with two or

more adverse SDOH, such as low income, food insecurity, unemployment, or lack of insurance, have a 1·5–2-fold higher likelihood of developing cardiometabolic MLTC.⁴⁰ This is compounded by neighbourhood-level deprivation, including poverty, poor health literacy, poor walkability, and rurality, which drive a four-fold higher cardiovascular event rate in socioeconomically disadvantaged areas.⁴¹ Environmental pollution further accelerates disease progression. Long-term exposure to fine particulate matter and nitrogen dioxide is independently associated with the transition from a single cardiometabolic condition to MLTC, with a hazard ratio of 1·03 (95% CI 1·01–1·06) and 1·06 (1·03–1·09) per interquartile increase.⁴²

Emerging quasi-experimental studies and natural experiments strengthen causal inference beyond observational associations. Studies exploiting compulsory schooling reforms in Europe have shown that additional years of education causally reduce cardiometabolic risk, independent of socioeconomic position.^{43,44} Within-sibling designs from Scandinavian registry-based cohorts, controlling for familial and genetic confounding, confirmed the independent effects of socioeconomic position on cardiometabolic outcomes.⁴⁵ Besides, policy-level natural experiments of urban low-emission zones and sugar-sweetened beverage taxes (such as in Mexico) link environmental and nutritional interventions to reduced cardiovascular events and metabolic benefits, despite limited long-term cardiometabolic MLTC-specific data.^{42,46,47} Real-world GLP-1 receptor agonist use also suggests sustained weight loss might attenuate progression towards MLTC, requiring long-term prospective validation.⁴⁸ Although these approaches remain susceptible to residual confounding, they provide a complementary hierarchy of evidence that moves beyond cross-sectional associations and strengthens the case for targeted SDOH-level interventions.

Mechanisms linking SDOH to cardiometabolic MLTC

Adverse SDOH contribute to the development and progression of cardiometabolic conditions through interconnected social, behavioural, and biological mechanisms (figure 1). These determinants shape the lifestyle patterns central to cardiometabolic risk. People with economic instability, low educational attainment, those from socioeconomically disadvantaged regions, or those with chronic exposure to environmental pollutants are more likely to adopt unhealthy behaviours, including suboptimal dietary patterns, physical inactivity, smoking, and excessive alcohol consumption.^{35,49,50} These behaviours reflect underlying structural constraints rather than personal preferences.^{50,51}

Access to healthy nutrition is constrained by the local food environment, particularly by the existence of food deserts (areas with limited access to affordable nutritious food) and food swamps (neighbourhoods saturated with energy-dense fast-food outlets).^{37–39} Educational

attainment mediates cardiometabolic risk not only through income and employment, but also through health literacy, which influences treatment adherence and preventive health behaviours.⁴³ Cultural norms also influence health-seeking behaviours; for instance, associating larger body size with prosperity and social status might delay obesity risk recognition and interventions.^{49,50} These upstream determinants of lifestyle are therefore priority targets for structural and policy-level interventions to reduce the cardiometabolic MLTC burden.⁵¹

The relative contribution of SDOH domains to cardiometabolic MLTC varies substantially across geographical and economic contexts. In low-income and middle-income countries (LMICs), rapid urbanisation, nutritional transition, environmental pollution (particularly household and ambient air pollution), and limited health-care infrastructure shape cardiometabolic trajectories.^{1,3,4} The double burden of malnutrition in these settings, where undernutrition and overnutrition increasingly coexist within the same populations and even within households, accelerates the progression from metabolic risk to MLTC.^{52–54} By contrast, in high-income countries, psychosocial stress, sedentary built environments, ultra-processed diets, and social isolation are more prominent mediators, although marginalised high-income country populations frequently mirror LMIC patterns.^{41,51} Notably, most epidemiological evidence originates from high-income country cohorts such as the UK Biobank,^{42,46} limiting generalisability to LMICs, where the MLTC burden is highest and increasing most rapidly.²

Chronic exposure to poverty, discrimination, and environmental stress activates neuroendocrine systems, particularly the sympathetic nervous system and the hypothalamic–pituitary–adrenal axis, leading to elevated catecholamines and glucocorticoids that promote insulin resistance, visceral adiposity, hypertension, and dyslipidaemia.^{51,55} Another driver of this neuroendocrine activation is the chronic psychosocial stress induced by varying cultural body norms and obesity stigma.⁵⁶ Sustained stress also drives systemic inflammation, reflected by increased concentrations of cytokines such as IL-6 and TNF and elevated high-sensitivity C-reactive protein, thereby worsening atherosclerotic risk and metabolic dysfunction.^{41,55,57} Glucocorticoid resistance further exacerbates inflammation, endothelial dysfunction, and vascular ageing.⁵⁵ Besides, chronic SDOH-related stress alters immune cell profiles and induces epigenetic changes, such as DNA methylation and histone acetylation, while accelerating telomere shortening and cardiometabolic risk.^{55,57,58} Emerging evidence suggests that even low-level, long-term exposure to environmental pollution perturbs glucose homeostasis and vascular integrity.⁵⁷

Taken together, SDOH are fundamental determinants of cardiometabolic MLTC risk, management, and health

outcomes with multiple adverse exposures increasing risk in a dose-dependent manner.^{35,40,41} Importantly, these impacts accumulate across the life course: childhood adversity becomes biologically embedded, driving adult adiposity, dyslipidaemia, and worsening cardiometabolic risk, independent of psychological or lifestyle factors in later life.^{59,60} Addressing current evidence gaps requires urgent, region-specific research and policy, particularly prospective cohort studies and intervention trials embedded within LMIC health-care systems, to inform context-appropriate prevention.^{3,6}

Role of early-life exposures

Although cardiometabolic MLTC are known to be driven by lifestyle factors, the underlying biology reflects complex interactions among social, nutritional, and environmental exposures. Herein, we explore how nutrition, the gut microbiome, and endocrine-disrupting chemicals (EDCs) contribute to MLTC.

Overnutrition and undernutrition

Both extremes of nutritional imbalance predispose adults to cardiometabolic MLTC, particularly in LMICs with rapid nutrition transitions.^{52,53} Undernutrition, arising from chronic energy insufficiency, inadequate protein intake, or micronutrient deficiencies, can induce metabolic and physiological adaptations that impair immune function and increase susceptibility to recurrent infections that can worsen tissue damage.⁵³ Specific micronutrient deficiencies drive additional cardiometabolic dysfunctions: iron deficiency impairs oxidative metabolism; vitamin B12 and folate deficiencies increase homocysteine concentrations and contribute to endothelial dysfunction; and zinc, magnesium, and vitamin D deficiencies impair insulin action and lipid metabolism.^{52,61} These challenges might be compounded by polypharmacy, as some medications for MLTC can suppress appetite or alter nutrient absorption. The thrifty phenotype hypothesis highlights the role of early-life nutrition.⁶² People exposed to limited nutrition in utero or during early childhood develop metabolic adaptations that favour energy conservation. Upon later exposure to calorie excess, they face an increased risk of insulin resistance, visceral adiposity, hypertension, and type 2 diabetes.^{61,62}

Overnutrition, often driven by excessive caloric intake, high consumption of ultra-processed foods, and sedentary lifestyles, promotes obesity and metabolic dysfunction, closely aligning with the shared mechanisms outlined previously.⁶³ Notably, overnutrition amplifies risk among people with early-life undernutrition, creating a double burden that helps explain the disproportionately high prevalence of type 2 diabetes and relevant cardiometabolic conditions in LMICs.⁵⁴ Although undernutrition and overnutrition are on opposite extremes, they converge through shared pathways involving nutritional imbalance, inflammation, and metabolic dysfunctions (figure 1).⁵³ This convergence

underscores how both loss of lean mass associated with undernutrition and excess adiposity arising from overnutrition (sometimes termed sarcopenic obesity) can create the interconnected risks, which lead to cardiometabolic MLTC.^{53,54,62}

Alterations in the gut microbiome

The gut microbiome represents a crucial interface through which early-life exposures can influence long-term susceptibility to cardiometabolic MLTC (figure 1). Both poor diet quality (overconsumption of ultra-processed foods) and early-life undernutrition are associated with alterations in the development of gut microbiota.⁶³ A healthy gut microbiome is characterised by high diversity and the presence of beneficial genera such as *Bifidobacterium* and *Lactobacillus*.⁶⁴ These bacteria produce short-chain fatty acids that can enhance insulin sensitivity, regulate appetite, maintain gut barrier integrity, and modulate inflammation.⁶⁵ By contrast, dysbiosis—often linked to diets low in fibre and high in refined carbohydrates, antibiotic exposure, or chronic stress—can reduce the production of short-chain fatty acids and increase intestinal permeability. The increase in intestinal permeability might enable microbial lipopolysaccharide to enter circulation, potentially triggering metabolic endotoxaemia. This persistent low-grade inflammatory state is implicated in obesity, MASLD, insulin resistance, and atherosclerosis, contributing to cardiometabolic MLTC.⁶⁵ Microbiome-derived metabolites add further complexity: TMAO (produced from dietary choline and red meat) has been associated with accelerated atherosclerosis and platelet activation, while the gut microbial regulation of bile acid metabolism appears to influence both glucose and lipid homeostasis.⁶⁶ Taken together, these pathways highlight how early-life alterations in the gut microbiome, when exacerbated by nutritional adversity across the life course, can increase susceptibility to cardiometabolic MLTC.

EDCs

Beyond nutritional adversity and gut microbiome alterations, EDCs are key contributors to obesity, dyslipidaemia, and cardiovascular disease, accelerating the development of cardiometabolic MLTC (ie, early-life programming box in figure 1).⁶⁷ Commonly found in plastics, pesticides, food packaging, cosmetics, and industrial emissions, EDCs are known to interfere with hormonal functions, even at low doses. EDCs such as bisphenol A, phthalates, polychlorinated biphenyls, dioxins, and organophosphate pesticides are lipophilic and accumulate in adipose tissue, contributing to chronic inflammation.⁶⁸ Pesticide exposure has been associated with increased risks of cardiovascular disease and metabolic syndrome.⁶⁸ These EDCs act through multifaceted biological pathways: (1) alterations of methylation patterns and gene expression involved in glucose and lipid metabolism, blood pressure regulation,

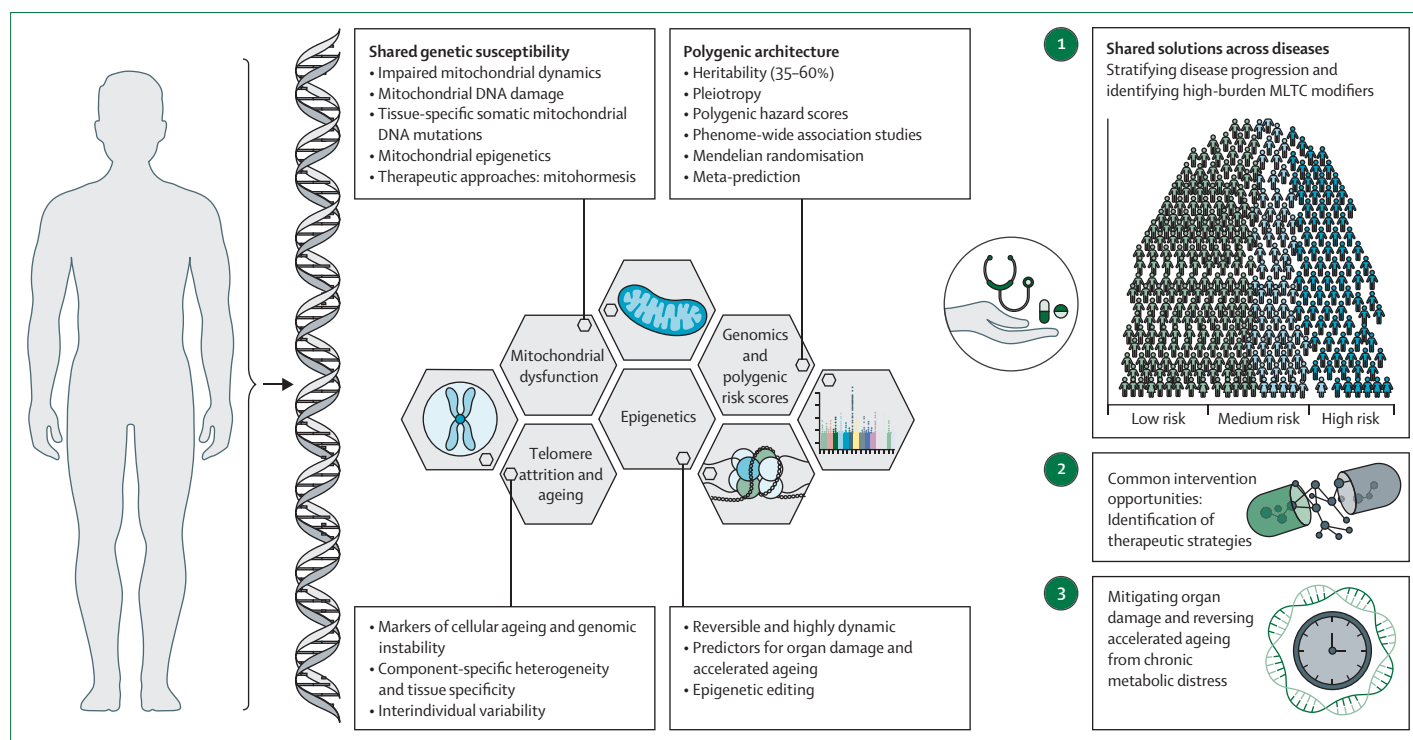


Figure 2: Shared genetic and epigenetic drivers of cardiometabolic MLTC and opportunities for multidisease therapeutic strategies

Elucidating shared molecular mechanisms enables a unified therapeutic approach across three primary domains: (1) shared solutions across diseases, focusing on stratifying disease trajectories and identifying high-burden MLTC modifiers; (2) common intervention opportunities, centred on the identification and development of broad-spectrum therapeutic strategies; and (3) organ protection, aiming to prevent downstream organ damage and reverse the accelerated ageing phenotypes induced by chronic metabolic distress. MLTC=multiple long-term conditions.

and chronic inflammation; (2) inhibition of thyroid hormone transport and metabolism, lowering metabolic rate, and promoting obesity; and (3) pathological activation of peroxisome proliferator-activated receptors (PPARs), particularly PPAR- γ .^{68–70} Unlike therapeutic PPAR- γ agonists, EDCs induce dysfunctional adipogenesis and off-target effects. By binding to other steroid or oestrogen receptors, EDCs drive systemic inflammation and insulin resistance. Early-life or chronic low-level exposure to EDCs exerts an obesogenic effect, characterised by irreversible adipocyte expansion and preferential visceral fat accumulation beginning in prenatal life, resulting in lifelong susceptibility to obesity and cardiometabolic MLTC.^{68–70}

The interactions between nutritional adversity across the life course, altered gut microbiome, and EDCs form a convergent pathway with increased susceptibility to cardiometabolic MLTC. These perturbations also imprint durable molecular effects, including altered genetic profiling and epigenetic modifications.⁷⁰

Role of genetic and epigenetic biomarkers

Early-life exposures represent a crucial window, in which adverse biological, SDOH, environmental, and lifestyle factors interact with underlying genetic susceptibility to influence epigenetic regulation. This interplay between early-life environments and molecular regulation

underscores the importance of genetic and epigenetic biomarkers in the development and progression of cardiometabolic MLTC (figure 2).

Genetic insights into cardiometabolic MLTC

MLTC is characterised by a polygenic and shared inheritance model. Susceptibility is associated with the cumulative, modest effects of multiple common genetic variants, which interact alongside rare and low-frequency variants that contribute to the genetic architecture of these complex traits.^{71–73} Depending on the specific combination of conditions, genetic factors are estimated to account for 35–60% of the phenotypic variability in MLTC. To facilitate clinical translation, this genetic evidence can be delineated into two primary functional domains: genomic risk stratification for personalised medicine and mechanistic validation for targeted therapeutic development.

Genetic risk stratification for personalised medicine

Genome-wide association studies (GWAS) findings have been leveraged to develop polygenic risk scores (PRS), which combine the effects of multiple genetic variants to improve population risk prediction and disentangle disease heterogeneity. For instance, recent PRS analyses identified at least two MASLD subtypes with distinct clinical trajectories: a liver-confined, aggressive form and

a systemic form associated with increased cardiometabolic risk.⁷⁴ Furthermore, novel integrative approaches, such as meta-prediction models incorporating 15 derived meta features alongside multiple embedded PRS, have generated a 10-year risk score for incident coronary artery disease (area under the curve 0·84; 95% CI 0·83–0·85), significantly outperforming established clinical models.⁷⁵ These findings address a central question in MLTC research: whether the cardiometabolic MLTC framework provides predictive utility beyond the mere summation of individual, condition-specific risk scores. These integrative models suggest that accounting for shared genetic architecture and systemic metabolic dysfunction offers superior risk stratification by capturing the latent, interconnected pathways that precede the clinical manifestation of discrete pathologies.^{75,76}

Moving beyond static risk estimates, polygenic hazard scores provide information on the timing of disease onset rather than lifetime risk alone. Using time-to-event methods, polygenic hazard scores can stratify people by progression patterns, linking genetic risk to age of onset and the development of specific comorbidities. This approach enables better prediction of cardiometabolic disease trajectories and identification of intervention opportunities.⁷⁷ Although promising, the clinical use of PRS and polygenic hazard scores is often constrained by the under-representation of diverse ancestries. Hence, developing genetic risk scores from multi-ancestry GWAS is crucial for equitable risk stratification.⁷⁸ Expanding on this concept, phenome-wide association studies improve the precision of genetic signals for core metabolic traits while broadening the detection of pleiotropic associations across the cardiometabolic MLTC spectrum.⁷⁹ Ultimately, the clinical translation of genetic risk stratification depends heavily on overcoming logistical barriers and securing professional acceptance within standard-of-care screening protocols.

Mechanistic validation for therapeutic development

The second core utility of genetic data lies in identifying and validating therapeutic targets through genotype–phenotype associations. In this context, insights from extreme phenotypes and rare, high-effect mutations have been instrumental in defining the biological boundaries of cardiometabolic health and clarifying how specific factors contribute to disease development. Genomic loci exhibiting pleiotropic effects suggest a genetic overlap among circulatory system conditions, such as coronary heart disease and atrial fibrillation.⁸⁰ However, a major challenge for clinical translation is prioritising causally relevant variants within these disease clusters, particularly when variants exert bidirectional or opposing effects. Although most genetic associations are directionally consistent, certain variants reveal complex biological trade-offs that mirror clinical trial evidence. For instance, variants in *HMGCR* and *NPC1L1* that lower LDL cholesterol are associated with a modest increase in

type 2 diabetes risk, providing genetic validation for the metabolic profiles observed with lipid-lowering therapies.⁸¹ Similarly, the *TM6SF2* Glu167Lys variant presents a therapeutic paradox: although it correlates with lipid traits⁸² and reduces myocardial infarction risk,⁸³ it simultaneously increases susceptibility to MASLD.^{82,84,85} This divergence in organ-specific effects, in which hepatic fat accumulation occurs at the expense of circulating lipids, illustrates a considerable challenge for therapeutic development. Pharmacological modulation of such pleiotropic pathways requires the careful balancing of cardiovascular protection against the risk of progressive liver injury.

Beyond risk stratification, PRS can help identify specific proteins implicated in the pathogenesis of cardiometabolic MLTC. In a population-based cohort, PRS for type 2 diabetes, coronary artery disease, ischaemic stroke, and CKD correlated with the levels of 49 plasma proteins, 12 of which represent candidate therapeutic targets.⁸⁶ Another cohort study that validated PRS explained up to 45·5% of the variance in 257 circulating protein biomarkers relevant to cardiometabolic conditions.⁸⁷ Multi-ancestry GWAS have also identified convergent genetic signals across cardiovascular–kidney–metabolic conditions, including variants in genes such as *LACTB* and *PLA2G6*.⁸⁸ *LACTB* plays an important role in cellular protection against ferroptosis—a regulated form of necrotic cell death. Building on this, the integration of MASLD into the cardiometabolic MLTC framework captures the broader spectrum of systemic metabolic dysfunction and supports more precise, cross-condition therapeutic approaches.⁷⁶

Finally, GWAS data are increasingly applied in mendelian randomisation studies to evaluate potential causal relationships. By leveraging genetic variants as natural instruments, mendelian randomisation aims to minimise confounding and reverse causation to infer true biological directionality. Using this approach, evidence suggests causal relationships between cardiometabolic risk factors and the development of MLTC.⁸⁹ Notably, mendelian randomisation analyses show that genetic susceptibility to type 2 diabetes is a primary determinant of 21 distinct conditions, including vascular dementia and polycystic ovarian syndrome, providing robust evidence for its major role in the pathogenesis of cardiometabolic MLTC.⁹⁰

Molecular ageing signatures

Although genetic biomarkers provide a roadmap of inherent risk, epigenetic biomarkers offer insights into the dynamic processes underlying cardiometabolic MLTC progression (figure 2). Given that current evidence largely derives from single-disease associations, further research is required to determine whether epigenetic mechanisms, such as DNA methylation and microRNAs, act as causal mediators or simply serve as molecular mirrors reflecting the interplay between biological risk

and SDOH.^{91–94} A multi-ancestry GWAS combined with single-cell epigenomics analysis has provided insights into the aetiology of type 2 diabetes by delineating the interplay of genomics and epigenomics.⁹⁵ The study resolved 1289 association signals (including 145 novel loci) into eight distinct clusters, each characterised by unique cardiometabolic associations and enrichment for cell type-specific open chromatin in tissues, such as pancreatic islets and adipocytes. Notably, cluster-specific PRS showed predictive ability for vascular outcomes across diverse ancestries, highlighting the role of obesity-related processes and polygenic heterogeneity in the pathogenesis of cardiometabolic conditions.⁹⁵

The proposed interplay among epigenetic alterations, telomere attrition, and mitochondrial dysfunction (as outlined in the previous section) is hypothesised to contribute to the molecular framework of cardiometabolic MLTC. Telomere length—a frequently cited biomarker of cellular ageing—shows significant associations with multiple conditions.⁹⁶ Shorter leukocyte telomere length is linked to an increased risk of type 2 diabetes,⁹⁷ coronary artery disease,⁹⁸ and CKD.⁹⁹ For instance, the paradoxical association between high LDL cholesterol and longer telomere length suggests that these biomarkers might reflect broader metabolic signalling rather than a unidirectional causal pathway.¹⁰⁰ Furthermore, whether telomere attrition directly induces cellular dysfunction across all relevant tissues (eg, adipose tissue) or serves as a surrogate marker of systemic stress remains unclear.¹⁰¹ Despite these mechanistic insights, defining the clinical use of leukocyte telomere length within the heterogeneous landscape of cardiometabolic MLTC remains a substantial translational challenge.

Emerging non-genetic biomarkers

Although genetic and epigenetic biomarkers offer important insights into upstream mechanisms, they capture only part of the biological landscape. Large-scale, high-dimensional studies have reported that circulating metabolites and fatty-acid signatures, as well as combined polygenic (as outlined in the previous section) and multiomics risk scores, provide complementary information on underlying physiological dysregulation in addition to established clinical models.¹⁰² Herein, we discuss the role of molecular biomarkers as observable windows into the shared mechanistic pathways, rather than serving as primary causal drivers. These emerging biomarkers hold promise for a more comprehensive understanding of disease trajectories, identifying high-risk phenotypes and refining risk stratification to facilitate effective precision prevention and therapeutic strategies.

High-throughput metabolomics, functioning as a single-domain, multi-disease assay across five population-based cohorts, has shown that signatures of amino acids, fatty acids, and energy metabolism (eg, carbohydrate and ketone bodies) alongside kidney function biomarkers

were associated with the incidence of 24 conditions, including cardiometabolic MLTC, dementia, and cancer, after a median follow-up of 12·2 years.¹⁰² These biomarkers improved individual-level risk discrimination beyond established clinical models.¹⁰² Notably, they have shared pathophysiological pathways such as insulin resistance, inflammation, and mitochondrial substrate alterations that underpin cardiometabolic MLTC.^{1,102} Using nuclear magnetic resonance spectroscopy in nearly 120 000 UK Biobank participants, a biomarker disease atlas comprising 249 metabolites and lipoproteins was developed. This atlas mapped robust associations across 700 conditions, showing the breadth of nuclear magnetic resonance-detectable biomarkers that could be scaled for population-level health assessment in people at high risk.¹⁰³

Similarly, high-throughput measurements from 52 000 UK Biobank participants were used to develop a disease-specific and mortality-specific proteomic risk score based on 1461 identified plasma proteins.¹⁰⁴ After a median follow-up of 14·1 years, the best predictive performance was achieved by combining an established clinical model (incorporating sociodemographic factors, lifestyle, comorbidity status, and medication history) with the proteomic risk score (which included GDF15, CDCP1, HAVCR1, and IL-6). This combined approach successfully stratified disease trajectories for cardiometabolic MLTC, which was particularly favourable within 5 years of disease onset, underscoring the substantial role of plasma proteomics in predicting early risk of disease progression.¹⁰⁴ Notably, GDF15 regulates endothelial adaptations following vascular damage, while cellular stress and mitochondrial dysfunction can drive its overexpression with detrimental effects on overall cardiometabolic health.^{105,106}

Among specific metabolomic signatures, branched-chain amino acids have emerged as one of the most reproducible signatures linking metabolic dysfunction to cardiometabolic conditions.^{107–109} Several cohort studies show that elevated circulating branched-chain amino acids are associated with incident type 2 diabetes, cardiovascular disease, and mortality.^{109–111} Mechanistically, chronic elevations of branched-chain amino acids appear to reflect impaired catabolism in peripheral tissues, linking them to perturbations in insulin signalling and myocardial metabolic inflexibility.^{108,110,111} Research is underway to investigate whether targeting the metabolism of branched-chain amino acids using either pharmacotherapies or modulation of gut microbiota could alter downstream trajectories of cardiometabolic MLTC.¹⁰⁸

Compared with single-layer biomarkers, multiomics integration can improve predictive performance for incident cardiometabolic conditions.^{104,112} In prospective analyses combining proteomics, metabolomics, and genomics, this multiomics model identified people at risk who were missed by routine clinical prediction algorithms.¹¹³ These algorithms included key risk factors

Panel: Key evidence gaps and future research priorities in cardiometabolic MLTC

Shared risk factors and mechanisms

- Identify shared causal molecular pathways for therapeutic targeting
- Delineate the neuroendocrine-immune axes that drive the development of cardiometabolic multiple long-term conditions (MLTC)
- Delineate the temporal stages in the evolution of cardiometabolic MLTC to identify critical windows for intervention

SDOH

- Standardise consensus-based criteria for the uniform definition of social determinants of health (SDOH) and objective measurements of cardiometabolic MLTC and disease severity
- Broaden the scope of SDOH by capturing socioeconomic, environmental, behavioural, and structural factors (eg, social support, housing quality, and neighbourhood safety) using standardised instruments
- Investigate the link between SDOH and cardiometabolic MLTC by conducting prospective real-world studies in diverse populations, including in marginalised groups and different health-care settings

Early-life exposures and programming

- Delineate the mechanisms by which early-life exposures alter the composition and functions of the gut microbiome
- Define the window-specific contribution of different developmental periods, from preconception, in utero, infancy, and adolescence to the final risk of cardiometabolic MLTC
- Analyse longitudinal trajectories for determining the crucial transition point at which cardiovascular and metabolic risk factors solidify into cardiometabolic MLTC

Genetic and epigenetic biomarkers

- Identify genetic modifiers within high-burden cardiometabolic MLTC, such as cardiovascular–kidney–liver–metabolic conditions
- Identify possible rare and structural variants and characterise their contributions to account for missing heritability
- Determine the relative influence of genetic, phenotypic, and environmental crosstalks on interindividual differences in MLTC progression
- Determine the cell-specific, persistent epigenetic changes (eg, DNA methylation and histone modifications in adipocytes or endothelial cells) that explain the effect of early-life exposures
- Measure and validate mitochondrial damage and dynamics markers (eg, cell-free mitochondrial DNA and key metabolites) in ethnically diverse cohorts

Emerging non-genetic biomarkers

- Integrate multiomics data for the discovery of novel, cross-disease biomarkers for improved risk stratification, prediction, and prognostication of cardiometabolic MLTC
- Validate multiomics signatures that can predict the transition from a single disease to cardiometabolic MLTC and the disease trajectories
- Delineate circulating molecular ageing markers that can predict and mediate inflammation-driven organ damage across various tissues

such as age, sex, glycated haemoglobin, BMI, lipid profile, and C-reactive protein.¹¹³ Compared with a standard diabetes risk score, a model comprising the top ten proteomic and metabolomic biomarkers improved predictive performance for incident type 2 diabetes from a C-index of 0·76 (95% CI 0·69–0·82) to 0·87 (0·82–0·92).¹¹³ These findings show that different molecular layers capture distinct aspects of pathophysiology and thus a

combined molecular approach can improve prognostication for the development and progression of cardiometabolic conditions.¹¹³

Current understanding of the polygenic heterogeneity of cardiometabolic MLTC is largely extrapolated from single-disease studies.^{114,115} This highlights a crucial gap, requiring explicitly designed, cross-ancestral, MLTC-focused genetic and multiomics research. From a translational perspective, several features make emerging biomarkers attractive for clinical implementation. First, many nuclear magnetic resonance-derived and mass spectrometry-derived biomarkers show high reproducibility and scalability for large-scale screening. These include targeted lipids and fatty acids profiling, branched-chain amino acid measurements, and now a single-domain, multi-disease assay.^{102,103} Second, people at risk can be stratified based on their dominant pathophysiological pathways, guided by the multiomics panels. This facilitates precision prevention, including tailored lifestyle interventions, targeted pharmacotherapy, and timely surveillance. Third, integrating these panels with PRS might enable early-life risk stratification, allowing multiomics biomarkers to be used for monitoring treatment responses and refining estimates of residual risk.^{113,114}

Despite these promises, several translational gaps persist. Because many existing studies are observational in design, robust causal inference methods are needed. Notably, mendelian randomisation analyses are required to distinguish whether biomarkers are merely correlates or true causal mediators that can be targeted by interventions. Standardisation of assay methods, cross-platform calibration, and demonstration of incremental clinical use beyond established clinical models (via net reclassification improvement and decision-curve analysis) are essential. Given the polygenic heterogeneity and varying baseline prevalences of cardiometabolic conditions, there is also an urgent need to validate these putative biomarkers across diverse populations.¹¹⁴ Finally, equitable access to these biomarkers is essential. Currently, high-throughput multiomics profiling is concentrated in well resourced settings, raising concerns that its rapid adoption could widen health disparities. This gap is particularly striking in LMICs, which face an increasing cardiometabolic MLTC burden but lack local replication of these multiomics studies. Strategic investment is urgently needed to prioritise populations at high risk, support cost-effective and locally adaptable approaches, and provide clear guidance on using these biomarkers across the cardiometabolic MLTC spectrum.¹¹³

Taken together, contemporary large-scale proteomic, metabolomic, and genomic studies provide a coherent view for a future precision health ecosystem: (1) baseline genomic profiling to identify genetic predisposition; (2) periodic and scalable biomarker screening to detect preclinical metabolic shifts; and (3) multiomics profiling to enhance prediction of incident cardiometabolic MLTC and disease progression.¹¹² Integrating this

comprehensive ecosystem into established clinical models and deploying an externally validated, single-domain, multi-disease risk score could enable earlier, more tailored prevention strategies to halt the progression from single-disease states to complex cardiometabolic MLTC. Large-scale, multinational, and ethnically diverse intervention studies are needed to highlight that such a biomarker-guided ecosystem can reduce hard clinical outcomes and is cost-effective across diverse health-care systems and resource settings.

Current gaps in knowledge and future research

Despite considerable progress, important gaps persist in delineating the mechanisms of cardiometabolic MLTC (panel). The temporal sequencing of biological perturbations, the causal hierarchy linking metabolic, inflammatory, and multi-organ signalling pathways, and the drivers of population-level heterogeneity are underexplored. A lack of longitudinal multiomics datasets, insufficient integration of environmental and sociobiological determinants, and the underrepresentation of diverse ancestries limit mechanistic inference. Addressing these gaps will require life-course cohort studies, deep phenotyping, and dynamic systems modelling to capture mechanistic transitions over time. Experimental and translational studies targeting shared pathways are essential to enhance precision prevention and therapeutics for cardiometabolic MLTC.

Conclusion

Cardiometabolic MLTC emerges from convergent perturbations across metabolic, inflammatory, and vascular pathways that collectively drive disease patterns and trajectories. By synthesising evidence from molecular pathways to multi-organ crosstalk, this Series paper delineates the mechanistic architecture underpinning systemic susceptibility and adverse health outcomes. These findings can identify novel opportunities for intervention and serve as a foundation for targeted prevention and management strategies that transcend single-disease frameworks (as outlined in the third paper in this Series).⁶ The continued integration of multiomics, systems biology, and life-course approaches will further enhance personalised risk profiling and therapeutic discovery. Ultimately, these advancements will allow us to address cardiometabolic MLTC as a truly interconnected system.

Contributors

KK conceived the idea and general plan for this Series. L-LL, AJ, DP, SS, RRB, and KK contributed content for the first draft of the manuscript. L-LL, AJ, DP, SS, RRB, and KK critically reviewed the final draft of the manuscript for important intellectual content and agreed to be accountable for all aspects of the manuscript. L-LL finalised the manuscript. All authors approved the final manuscript for publication.

Declaration of interests

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